

MANCHIKANTI SURYA VARDHAN

Hyderabad, India | +91-9676810809 | msuryavar@gmail.com |
[linkedin.com/in/manchikanti-surya-varadhan-485581211](https://www.linkedin.com/in/manchikanti-surya-varadhan-485581211) | github.com/msuryavardhan
Portfolio: [Portfolio Website](#)

Professional Summary

M.Tech VLSI Design student with hands-on experience in RTL design, functional verification, ASIC implementation, and custom IP core development. Proficient in Verilog HDL, Cadence Genus, Innovus, NCLaunch, and Xilinx Vivado, with experience in RTL-to-GDSII flow, logic synthesis, physical design, timing analysis, and FPGA-based implementation. Strong understanding of Digital IC Design, CMOS VLSI Design, FPGA Architecture, and RISC-V Processor Design. Seeking opportunities in RTL Design, Physical Design or Design Verification.

Technical Skills

HDL / Languages	Verilog HDL, SystemVerilog (learning), TCL, Linux Shell Scripting
EDA Tools	Cadence Genus, Cadence Innovus, Cadence NCLaunch, QuestaSim, ModelSim, Vivado, Vitis AI, Vitis HLS, Quartus Prime
ASIC Flow	RTL-to-GDSII Flow, Logic Synthesis, Floorplanning, Placement, CTS, Routing, STA
FPGA / SoC	Xilinx ZCU104 MPSoC, IP Integrator, DPU Integration, FPGA Prototyping, Altera DE10
Verification	Functional Simulation, Testbench Development, Waveform Debugging, SystemVerilog Assertions (SVA Basics)
Architecture	RISC-V (RV32I), MAC Unit Design, FIR Filter Design, SoC/IP Integration
Core Concepts	Digital Logic Design, CMOS VLSI Design, Digital IC Design, FPGA Architecture, ASIC Design Flow, Physical Design

Projects

32-bit RISC-V Processor with Integrated MAC Unit — RTL-to-GDSII ASIC Flow

- Designed and implemented a 32-bit single-cycle RISC-V processor with an integrated MAC Unit using Verilog HDL.
- Executed the complete ASIC Physical Design flow including Floorplanning, Power Planning, Placement, Clock Tree Synthesis (CTS), Routing and Sign-off.
- Performed logic synthesis using Cadence Genus and physical implementation using Cadence Innovus with gpdk90 (90nm) technology library.
- Applied SDC timing constraints and analyzed setup and hold timing violations throughout implementation stages.
- Performed timing optimization across placement, CTS and post-route stages to achieve timing closure.
- Generated utilization, congestion, timing and power reports for design quality assessment.

FPGA-Based Drone Detection and Classification System using YOLOv8n

- Implemented a hardware-accelerated drone detection pipeline on FPGA using lightweight **YOLOv8n** architecture for real-time edge AI inference.
- Trained and fine-tuned the YOLOv8n model on a custom drone dataset, improving detection precision by approximately **5.3%**.
- Exported the trained model to **ONNX** format, performed INT8 quantization using Vitis AI, and deployed it on the **Xilinx ZCU104 MPSoC** using DPU acceleration.
- Designed and integrated a custom IP Core for Non-Maximum Suppression (NMS) using Vivado HLS, enabling hardware-accelerated post-processing of object detections.

VLSI Design of 2D FIR Filter Architecture using CSD

- Designed a 2D FIR Filter Architecture using **Canonical Signed Digit (CSD)** representation to eliminate conventional multipliers.
- Implemented multiplier-less DSP architecture using optimized shift-and-add operations, reducing hardware complexity and power consumption.
- Generated and optimized FIR filter coefficients using MATLAB and implemented the architecture using Verilog HDL in Xilinx Vivado.
- Achieved up to 17% reduction in area and 8% reduction in delay compared with the baseline 2D FIR filter

architecture.

Industrial Training & Internship

VLSI-Based IC DesignMay 2026 – Jul 2026

Academy of Skill Development

- Completed an **8-week industrial training program** focused on VLSI-based IC design and implemented an **8-bit ALU using RISC-V**.
- Applied RTL design concepts and digital logic principles to develop and verify the ALU architecture as part of the training project; awarded **A++** upon completion.

Education

Vellore Institute of Technology, Chennai2025 – 2027

Master of Technology (M.Tech) in VLSI Design | CGPA: 8.45

CVR College of Engineering, Hyderabad2019 – 2023

Bachelor of Technology (B.Tech) in
Electronics and Communication Engineering | CGPA: 8.23

Narayana Junior College, Hyderabad2017 – 2019

Intermediate (MPC) | Percentage: 89.9%

Narayana E-Techno School, Hyderabad2017

SSC (10th Grade) | CGPA: 9.3

Relevant Coursework

- Physical Design
- Static Timing Analysis
- ASIC Design Flow
- Digital IC Design
- CMOS VLSI Design
- Verification Methodologies
- FPGA Architecture

Certifications

- Semiconductor Skill Accelerator (SSA) Program** — IIT Bombay & VLSI System Design (VSD) 2026
- Workshop on RTL-to-GDSII Physical Design Flow** — ProV LogicJun 2026
- Workshop on VLSI Design & Semiconductor Industry** — ChipXpert Technologies Pvt. Ltd. May 2026
- ASIC Flow, Logic Synthesis, STA, Physical Design** — Cadence Genus, Tempus, Innovus & Synopsys ICC2 (Udemy - VLSI Mentor)Ongoing
- Certified in CMOS Digital VLSI Design** — NPTEL / SWAYAM2025
- Certified in Verilog HDL — Hands-On** — Maven Silicon2025